Project Initialization and Planning Phase

Date	23 June 2026
Team ID	SWTID-2026-5853
Project Name	A Comprehensive Measure of Well-Being (Human Development Index Prediction)
Maximum Marks	3 Marks

Project Proposal (Proposed Solution) report

Section	**Details**
Objective	Develop a machine learning web application to predict the Human Development Index (HDI) category using Life Expectancy, Mean Years of Schooling, Expected Years of Schooling and GNI per Capita.
Scope	Provide HDI prediction, country comparison, data visualization and downloadable reports for researchers and policymakers.
Problem Description	Manual HDI assessment is time-consuming and requires analysis of multiple socio-economic indicators.
Impact	Improves decision-making, reduces analysis time and supports evidence-based development planning.
Approach	Use Python, Flask and Scikit-learn to preprocess data, train an ML model and predict HDI categories.
Key Features	• HDI category prediction • Interactive dashboard • Visual charts • Country comparison • Report generation

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware	Processor	Intel Core i3 or above
Hardware	Memory	Minimum 4 GB RAM
Hardware	Storage	10 GB Free Space
Software	Framework	Flask
Software	Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Software	Development Environment	Anaconda, Jupyter Notebook / Spyder
Data	Dataset	HDI dataset (CSV)